

Introduction to Machine Learning

Study Notes — Chapter 1 & 2

Overview

Machine Learning (ML) is a subset of Artificial Intelligence (AI) that allows systems to learn and improve from experience without being explicitly programmed. ML focuses on developing computer programs that can access data and use it to learn for themselves. The term was coined by Arthur Samuel in 1959.

Types of Machine Learning

1. Supervised Learning

The algorithm is trained on labeled data — input-output pairs. The model learns to map inputs to outputs. Examples include Linear Regression, Logistic Regression, Support Vector Machines (SVM), and Neural Networks. Used for: spam detection, image classification, price prediction.

2. Unsupervised Learning

The algorithm is trained on unlabeled data and must find hidden structure on its own. Examples include K-Means Clustering, DBSCAN, Principal Component Analysis (PCA), and Autoencoders. Used for: customer segmentation, anomaly detection, dimensionality reduction.

3. Reinforcement Learning

An agent learns to make decisions by interacting with an environment and receiving rewards or penalties. Key terms: Agent, Environment, State, Action, Reward. Examples include Q-Learning and Deep Q-Networks (DQN). Used for: game playing (AlphaGo), robotics, self-driving cars.

Key Formulas and Equations

Mean Squared Error (MSE): $MSE = (1/n) * \sum((y_i - \hat{y}_i)^2)$

Sigmoid Activation Function: $\sigma(x) = 1 / (1 + e^{(-x)})$

Gradient Descent Update Rule: $\theta = \theta - \alpha * \text{grad}_J(\theta)$, where α is the learning rate

Bayes Theorem: $P(A|B) = P(B|A) * P(A) / P(B)$

F1 Score: $F1 = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$

Bias-Variance Tradeoff

A fundamental concept in ML. High bias leads to underfitting — the model is too simple to capture the underlying pattern. High variance leads to overfitting — the model memorizes training data and fails to generalize. $\text{Total Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Noise}$.

Model Evaluation Metrics

$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$

$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$

$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$

Likely Exam Topics

- Differences between supervised, unsupervised, and reinforcement learning
- When to use classification vs. regression
- How gradient descent minimizes a cost function
- Overfitting vs. underfitting and remedies: regularization, dropout, cross-validation
- The purpose and formula of the sigmoid function in logistic regression
- K-Fold cross-validation and why it matters
- The role of the learning rate (alpha) in training convergence
- Precision vs. Recall vs. F1 Score and when to prioritize each